Predicting League of Legends Match Outcomes Using Team Compositions

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Abstract

Predicting match outcomes in League of Legends is a challenging problem with practical applications in gaming analytics. This project explores the feasibility of predicting the winning team using only champion compositions, employing machine learning techniques like Random Forests and Neural Networks. Results indicate modest predictive accuracy, with significant room for improvement through richer feature engineering and advanced modeling. This paper outlines the dataset, methodology, experiments, and future research directions for improving prediction accuracy.

1 Introduction

The popularity of multiplayer online battle arena (MOBA) games like League of Legends has spurred interest in predictive analytics for esports. Understanding which factors contribute to match outcomes can benefit players, analysts, and developers alike. This project aims to address the specific problem of predicting match outcomes based solely on team compositions, focusing on preprocessing techniques, model experimentation, and accuracy improvements. We present our dataset, methodology, experiments, and conclusions in this report.

League of Legends matches involve two teams of five champions, each filling specific roles: Top, Jungle, Mid, ADC (Attack Damage Carry), and Support. While match outcomes depend on in-game decisions, mechanical skill, and teamwork, this project isolates the impact of champion selection. Our key question is: How accurately can we predict match outcomes with only team composition as input?

Prediction models in MOBA games also have applications in coaching, matchmaking systems, and automated game commentary. A well-performing model could offer teams insights into optimal drafts or identify weak compositions before matches begin. The potential benefits extend to the broader esports ecosystem, making this a valuable area for study.

2 Background

Predicting esports outcomes has been explored in prior research, often emphasizing in-game statistics like gold lead, kill-death ratios, or turret destruction. However, few studies focus solely on pre-game factors like champion selection. This project builds on existing literature by simplifying the problem space, leveraging machine learning techniques to analyze the predictive power of champion compositions.

Champion roles significantly affect gameplay dynamics. For instance, some champions specialize in specific roles, while others are flexible. This variability adds complexity to predictions, necessitating robust preprocessing and feature engineering. In the esports domain, drafting strategies often hinge on meta trends, counter-picks, and synergies, further complicating predictions.

A unique aspect of this work is its exclusive focus on champion composition. Unlike many esports prediction models that incorporate dynamic in-game statistics, we deliberately omit these variables to study the isolated impact of pre-game choices. This approach highlights both the challenges and potential of composition-based predictions.

3 Data

Our dataset consists of approximately 10,000 matches from various ranked tiers in League of Legends, collected using Riot Games' API. Each record contains:

- Match ID
- Champion compositions for both teams (five champions per team)
- Role assignments: Top, Jungle, Mid, ADC, Support
- Winning team (binary: 1 for Team 1, 0 for Team 2)

Preprocessing: The raw data required significant preprocessing to ensure usability. Teams were parsed into individual roles, and missing or invalid entries were filled with placeholder values. A LabelEncoder was employed to convert categorical values into numeric representations suitable for machine learning models.

The preprocessing pipeline included steps such as deduplication, role-based mapping, and handling unseen champions. Additionally, the dataset was balanced to ensure an equal distribution of Team 1 and Team 2 victories, minimizing the risk of model bias.

4 Methods

We experimented with multiple machine learning models, including:

- **Random Forest Classifier:** A robust ensemble model that handles categorical data effectively.
- **XGBoost:** A gradient-boosting framework optimized for speed and performance.
- **Shallow Neural Network:** A feedforward network with two hidden layers for nonlinear decision boundaries.

For feature engineering, champion names were encoded numerically, preserving distinctions across roles. The final feature set comprised ten input variables (five roles per team) and one binary target variable (winning team). Hyperparameter tuning was performed using grid search for Random Forests and Bayesian optimization for XGBoost.

Shallow Neural Network Architecture: The neural network used consisted of:

- Input layer: 10 neurons corresponding to the encoded features.
- Hidden layers: Two fully connected layers with ReLU activations (16 and 8 neurons, respectively).
- Output layer: A single sigmoid unit for binary classification.

Training was conducted over 50 epochs using the Adam optimizer and binary cross-entropy loss. Dropout layers were introduced to mitigate overfitting.

5 Experiments

The dataset was split into 80% training and 20% testing subsets. Cross-validation was used to ensure generalizability. Table 1 summarizes model performance.

Table 1: Model performance comparison.

Model	**Accuracy**	**F1-Score**
Random Forest	52.8%	0.53
XGBoost	53.2%	0.54
Shallow Neural Network	50.1%	0.47

The Random Forest model slightly outperformed others, likely due to its ability to capture complex interactions among features. However, all models fell short of the 60-70% accuracy hypothesized as the upper bound for champion composition-based predictions.

Error Analysis: Examination of misclassified matches revealed patterns such as:

- Teams with unconventional compositions (e.g., three support champions).

- Champions with multi-role flexibility, leading to ambiguity in role assignments.
- Matches involving less common champions, which the model struggled to encode effectively.

6 Conclusion

This project highlights the potential and limitations of predicting match outcomes using champion compositions. While the models achieved modest accuracy, results indicate the need for richer contextual data, such as player skill levels, in-game metrics, or champion synergies.

Future work could explore:

- Incorporating in-game statistics to complement pre-game features.
- Testing advanced architectures like Transformers for sequence-based champion encoding.
- Expanding the dataset to include professional matches for higher-quality data.

Ultimately, this study demonstrates that champion compositions are a meaningful, albeit incomplete, predictor of match outcomes in League of Legends.

References

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